

MA 310: Matrix Computations & Lab

Course Code	MA 310
Course Name	Matrix Computations & Lab
L-T-P-C	3-0-2-4
Prerequisites	Linear Algebra, Any programming language
Intended for	B.Tech. Mathematics & Computing
Distribution	Core for B. Tech. Mathematics and Computing, Elective for other disciplines
Mutual Exclusion	DS 402, DS 412
Approval	59th BoA (Apr 2025)

Course Contents

- **Fundamentals of Matrix Computations:** Overview of matrix computations, matrix norms (Frobenius, induced, etc.), stability and conditioning related to matrix computation and their effects, introduction to perturbation theory. [6 Lectures]
- **Linear Systems and Factorization Techniques:** LU factorization, Gaussian elimination, Cholesky factorization, and their sensitivity analysis. Iterative methods for solving linear systems: Jacobi, Gauss-Seidel, and Successive Over-Relaxation (SOR) methods. Stability and sensitivity analysis. [7 Lectures]
- **Linear Least Squares and QR Factorization:** Gram-Schmidt process, rotators, reflectors, QR factorization, stability in least squares problems, normal equations, and Moore-Penrose pseudoinverse. Sensitivity analysis in rank-deficient least squares problems. [8 Lectures]
- **Eigenvalues and Eigenvectors:** Spectral theorem for symmetric matrices, eigenvalue decomposition, Jacobi method, power and inverse power methods, Rayleigh quotient iteration, Schur's decomposition. Sensitivity analysis of eigenvalues and eigenvectors. [10 Lectures]
- **Singular Value Decomposition (SVD) and Applications:** Singular Value Decomposition (SVD), its applications in solving linear systems and rank-deficient problems, QR algorithm for SVD, error propagation in eigenvalue and singular value computations Stability and computational aspects of SVD. [11 Lectures]

Laboratory Modules

The computer lab classes will complement the theoretical understanding of the students. All the concepts, discussed in the lecture classes, will be used to solve practical problems through computer programming during the lab classes. This will provide a hands-on experience to the students on this subject.

Text Books

1. Watkins, D. S., **Fundamentals of Matrix Computations**, Vol. 64, John Wiley & Sons, 2004.
2. Golub, G. H., & Van Loan, C. F., **Matrix Computations**, 4th Edition, Johns Hopkins University Press, 2013.

References

1. L. N. Trefethen and David Bau, **Numerical Linear Algebra**, SIAM, 1997.
2. Demmel, J. W., **Applied Numerical Linear Algebra**, SIAM, 1997.
3. B. N. Datta, **Numerical Linear Algebra and Applications**, 2nd Edition, SIAM, 2010.
4. Gentle, J. E., **Matrix Algebra: Theory, Computations and Applications in Statistics**, Springer Texts in Statistics, 2017.